



Yakeen Plus (2025)

Practice Test - 14

DURATION : 200 Minutes

DATE : 12/12/2024

M. MARKS : 720

Topics Covered

Physics :	Physics And Measurement, Kinematics, Laws Of Motion, Work, Energy, and Power, Rotational Motion
Chemistry :	Some Basic Concept of Chemistry, Solutions, Organic Chemistry: Some Basic principles and Techniques, (IUPAC Naming), Organic Chemistry: Some Basic, principles and Techniques, (Isomerism), Organic Chemistry: Some Basic principles and Techniques, (GOC), Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure
Biology :	(Botany): Cell - The Unit of Life, Cell Cycle and Cell Division, The Living World, Biological Classification, Plant Kingdom (Zoology): Structural Organization in Animals, Animal Kingdom, Biomolecules and Breathing and Exchange of Gases

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Botany**) & Section IV (**Zoology**) and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

SECTION-(I) PHYSICS

SECTION - A

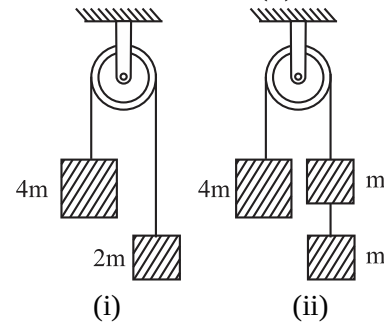
- Acceleration of a particle moving in x-y plane varies with time t as
 $\vec{a} = (\hat{i} + 3t^2 \hat{j})$
 Here a is in m/s^2 and t in second. At time $t = 0$, particle is at rest at origin. Mass of the particle is 1 kg. Find the net work done on the particle in first 2 s.
 (1) 40 J (2) 34 J
 (3) 16 J (4) 48 J
- Choose the **wrong** statement:
 (1) a dimensionally correct equation may be correct.
 (2) a dimensionally incorrect equation must be incorrect.
 (3) a dimensionally correct equation may be incorrect.
 (4) a dimensionally incorrect equation may be correct.
- Vander Waal's gas equation is:
 $\left(P + \frac{a}{V^2}\right)(V - b) = RT$. The dimensions of constant a as given above are:
 (1) $[\text{ML}^4\text{T}^{-2}]$ (2) $[\text{ML}^5\text{T}^{-2}]$
 (3) $[\text{ML}^3\text{T}^{-2}]$ (4) $[\text{ML}^2\text{T}^{-2}]$
- The velocity u of particles is given in terms of time t by the equation $u = at + \frac{b}{t^2 + c}$.
 The dimension of a , b and c are:
 (1) $[\text{L}^2]$, $[\text{T}]$, $[\text{LT}^2]$
 (2) $[\text{LT}^2]$, $[\text{LT}]$, $[\text{L}]$
 (3) $[\text{LT}^{-2}]$, $[\text{LT}]$, $[\text{T}^2]$
 (4) $[\text{L}]$, $[\text{LT}]$, $[\text{T}^2]$
- A force F is given by $F = at + bt^2$, where t is time. What are the dimensions of a and b ?
 (1) $[\text{MLT}^{-3}]$ and $[\text{ML}^2\text{T}^{-4}]$
 (2) $[\text{MLT}^{-3}]$ and $[\text{MLT}^{-4}]$
 (3) $[\text{MLT}^{-1}]$ and $[\text{MLT}^0]$
 (4) $[\text{MLT}^{-4}]$ and $[\text{MLT}^1]$
- Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: When the units of measurement of a quantity are changed, its numerical value changes.
Reason R: Smaller the unit of measurement, smaller is its numerical value.
 In the light of the above statements, choose the **correct** answer from the options given below:
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.

7. Match List-I with List-II.

	List-I		List-II
(A)	Surface tension	(I)	$\text{Kg m}^{-1} \text{s}^{-1}$
(B)	Pressure	(II)	Kg m s^{-1}
(C)	Viscosity	(III)	$\text{Kg m}^{-1} \text{s}^{-2}$
(D)	Impulse	(IV)	Kg s^{-2}

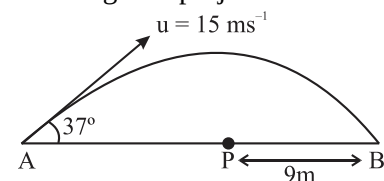
Choose the **correct** answer from the option given below:

- (1) A-IV, B-III, C-II, D-I
 - (2) A-IV, B-III, C-I, D-II
 - (3) A-III, B-IV, C-I, D-II
 - (4) A-II, B-I, C-III, D-IV
- In the arrangement shown in figure, pulley is smooth and massless and all the strings are light. Let F_1 be the force exerted on the pulley in case (i) and F_2 the force in case (ii). Then,



- (1) $F_1 > F_2$ (2) $F_1 < F_2$
- (3) $F_1 = F_2$ (4) $F_1 = 2F_2$

- A ball is hit by a batsman at an angle of 37° as shown in figure. The man standing at P should run at what minimum velocity so that he catches the ball before it strikes the ground. Assume that height of man is negligible in comparison to maximum height of projectile.



- (1) 3 ms^{-1} (2) 5 ms^{-1}
- (3) 9 ms^{-1} (4) 12 ms^{-1}

- In a new system of units, unit of mass is x kg, unit of length is y metre and unit of time is z second. Now if 1 newton = F new units, then $F =$

- (1) $\frac{z}{xy}$ (2) $\frac{z^2}{xy}$
- (3) $\frac{z}{xy^2}$ (4) $\frac{z}{x^2y}$

11. A body moves along curved path of a quarter circle. The ratio of magnitude of displacement to distance is:

(1) $\frac{\pi}{2\sqrt{2}}$ (2) $\frac{\pi}{2}$
 (3) $\frac{2\sqrt{2}}{\pi}$ (4) $\frac{3\pi}{2\sqrt{2}}$

12. In an experiment four quantities a , b , c and d are measured with percentage error 1%, 2%, 3% and 4% respectively. A quantity P is calculated as follows: $P = \frac{a^3 b^2}{cd}$, then % error in P is;

(1) 10% (2) 7%
 (3) 4% (4) 14%

13. In a vernier calipers, $(N + 1)$ divisions of vernier scale coincide with N divisions of main scale. If 1 MSD represents 0.1 mm, the vernier constant (in cm) is:

(1) $\frac{1}{10N}$ (2) $\frac{1}{100(N+1)}$
 (3) $100N$ (4) $10(N+1)$

14. Match the List-I with List-II.

List-I		List-II	
(A)	Angular momentum	(I)	$[ML^2T^{-2}]$
(B)	Torque	(II)	$[ML^{-2}T^{-2}]$
(C)	Stress	(III)	$[ML^2T^{-1}]$
(D)	Pressure gradient	(IV)	$[ML^{-1}T^{-2}]$

Choose the **correct** answer from the options given below.

- (1) A-I, B-IV, C-III, D-II
 (2) A-III, B-I, C-IV, D-II
 (3) A-II, B-III, C-IV, D-I
 (4) A-IV, B-II, C-I, D-III

15. A stone dropped from the top of the tower touches the ground in 4 sec. The height of the tower is about: ($g = 10 \text{ m/s}^2$)

(1) 80 m (2) 40 m
 (3) 20 m (4) 160 m

16. A ball is released from certain height h reaches ground in time T . Where will it be from the ground at time $3T/4$?

(1) $\frac{9h}{16}$ (2) $\frac{7h}{16}$
 (3) $\frac{3h}{4}$ (4) $\frac{27h}{64}$

17. Two bodies, A (of mass 1 kg) and B (of mass 3 kg) are dropped from heights of 16 m and 25 m, respectively. The ratio of the time taken by them to reach the ground is:

(1) 5/4 (2) 12/5
 (3) 5/12 (4) 4/5

18. A particle starts moving from rest with uniform acceleration. It travels a distance x in first 2s and distance y in the next 2s. Then:

(1) $y = 3x$ (2) $y = 4x$
 (3) $y = x$ (4) $y = 2x$

19. If a_r and a_t represents radial and tangential accelerations, the motion of a particle will be uniformly circular if:

(1) $a_r = 0$ and $a_t = 0$ (2) $a_r = 0$ but $a_t \neq 0$
 (3) $a_r \neq 0$ but $a_t = 0$ (4) $a_r \neq 0$ and $a_t \neq 0$

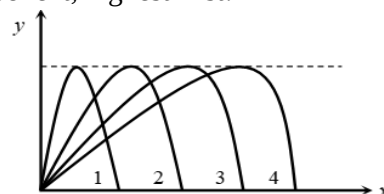
20. A boat is moving with velocity of $3\hat{i} + 4\hat{j}$ in river and water is moving with a velocity of $-3\hat{i} - 4\hat{j}$ with respect to ground. Relative velocity of boat with respect to water is:

(1) $-6\hat{i} - 8\hat{j}$ (2) $6\hat{i} + 8\hat{j}$
 (3) $8\hat{i}$ (4) $6\hat{i}$

21. Equation of motion of projectile in $x-y$ plane is $y = ax - bx^2$, then the angle of projection with x -axis is:

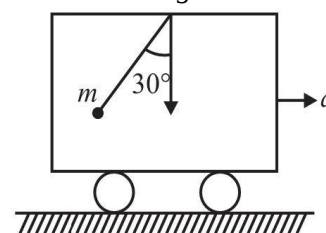
(1) $\tan^{-1}(b)$ (2) $\tan^{-1}(a)$
 (3) $\cos^{-1}(b)$ (4) $\sin^{-1}(a)$

22. Figure shows four paths for a kicked football. Ignoring the effects of air on the flight, rank the paths according to initial horizontal velocity component, highest first:



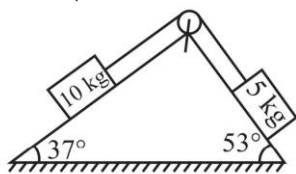
(1) 1, 2, 3, 4 (2) 2, 3, 4, 1
 (3) 3, 4, 1, 2 (4) 4, 3, 2, 1

23. If trolley accelerates horizontally with acceleration a , then bob is displaced backward from its initial vertical position by an angle 30° in equilibrium w.r.t. trolley as shown in figure. The value of a in terms of g is:

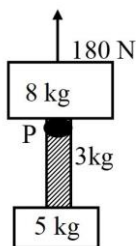


(1) g (2) $\sqrt{3}g$
 (3) $\frac{g}{\sqrt{3}}$ (4) $\frac{g}{2}$

24. The tension in the string as shown in the figure is: $[g = 10 \text{ m/s}^2]$ (All surfaces are smooth)

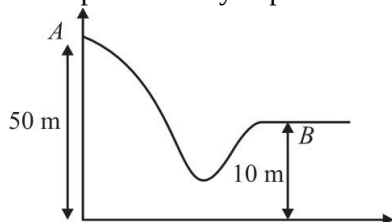


- (1) 23.3 N (2) 50 N
(3) 30 N (4) 46.67 N
25. Two blocks of mass 8 kg and 5 kg are connected by a heavy rope of mass 3 kg. An upward force of 180 N is applied as shown in the figure. The tension in the string at point P will be:

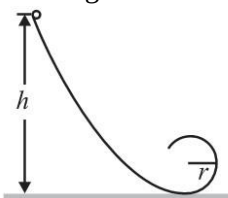


- (1) 60 N (2) 90 N
(3) 120 N (4) 150 N
26. Under the action of a force, a 1 kg body moves, such that its position x as a function of time t is given by $x = \frac{t^3}{2}$, where x is in meter and t is in second. The work done by the force in first 3 second is:
- (1) 243 J (2) 2430 J
(3) $\frac{729}{8}$ J (4) 24.3 J

27. A body is released from position A as shown in figure. The speed of body at position B is:



- (1) 10 m/s (2) $10\sqrt{2}$ m/s
(3) 20 m/s (4) $20\sqrt{2}$ m/s
28. A block follows the path as shown in the figure from height h . If radius of circular path is r , then relation that holds good to complete full circle is:



- (1) $h < 5r/2$ (2) $h > 5r/2$
(3) $h = 5r/2$ (4) $h \geq 5r/2$

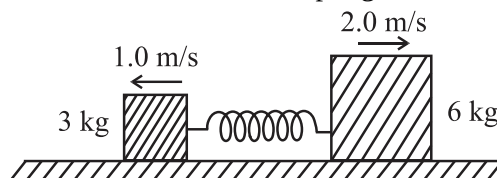
29. The potential energy of a particle in a force field is $U = \frac{A}{r^2} - \frac{B}{r}$ where A and B are positive constants and r is the distance of the particle from the centre of the field. For stable equilibrium, the distance of the particle is:

- (1) $\frac{B}{2A}$ (2) $\frac{2A}{B}$
(3) $\frac{A}{B}$ (4) $\frac{B}{A}$

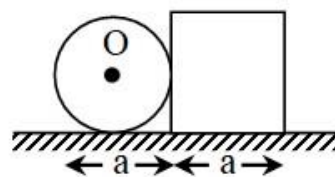
30. The position vector of three particles of masses $m_1 = 1 \text{ kg}$, $m_2 = 2 \text{ kg}$ and $m_3 = 4 \text{ kg}$ are $\vec{r}_1 = (\hat{i} + \hat{j})m$, $\vec{r}_2 = (2\hat{i} + 5\hat{j})m$ and $\vec{r}_3 = (2\hat{i} - \hat{j})m$ respectively. The position vector of their centre of mass is:

- (1) $(7\hat{i} - 13\hat{j})m$ (2) $\left(\frac{13\hat{i} + 17\hat{j}}{7}\right)m$
(3) $\left(\frac{13\hat{i} + 7\hat{j}}{7}\right)m$ (4) $(13\hat{i} + 17\hat{j})m$

31. Two blocks of mass 3 kg and 6 kg respectively are placed on a smooth horizontal surface. They are connected by a light spring of force constant $k = 200 \text{ N/m}$. Initially the spring is unstretched. The indicated velocities are imparted to the blocks. The maximum extension of the spring will be

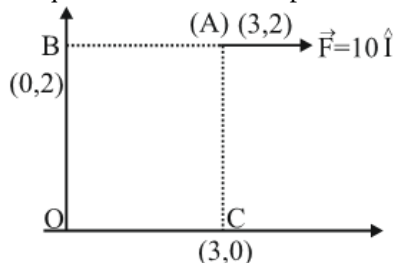


- (1) 30 cm (2) 25 cm
(3) 20 cm (4) 15 cm
32. A square plate of edge a and a circular disc of diameter a are placed touching each other at the midpoint of an edge of the plate as shown in figure. Locate the centre of mass of the combination, assuming same mass per unit area for the two plates.



- (1) $\frac{4a}{\pi + 4}$ (2) $\frac{a}{\pi + 4}$
(3) $\frac{4}{\pi + 3}$ (4) None of these

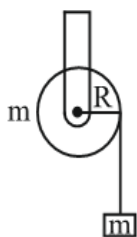
33. Find Torque of force F about point A and point C



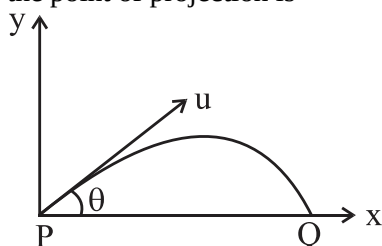
- (1) 0, 20 Nm (2) 10, 30 Nm
(3) 30, 0 Nm (4) 10, 10 Nm
34. A Uniform Rod of mass m and length L is shown in figure. Find α (angular acceleration) on releasing the rod.



- (1) $\frac{3g}{2L}$ (2) $\frac{6g}{5L}$
(3) $\frac{3g}{L}$ (4) $\frac{7g}{11L}$
35. A mass ' m ' is supported by a massless string wound around a uniform hollow cylinder of mass m and radius R . If the string does not slip on the cylinder, with what acceleration will the mass fall on release?

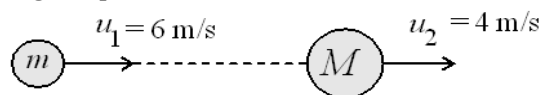


- (1) $\frac{5g}{6}$ (2) g
(3) $\frac{2g}{3}$ (4) $\frac{g}{2}$
- SECTION - B**
36. Average torque on a projectile of mass m , initial speed u and angle of projection θ between initial and final positions P and Q as shown in figure about the point of projection is

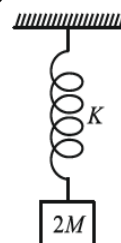


- (1) $\frac{mu^2 \sin 2\theta}{2}$ (2) $mu^2 \cos \theta$
(3) $mu^2 \sin \theta$ (4) $\frac{mu^2 \cos \theta}{2}$

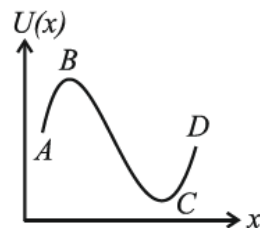
37. A particle of mass m moving with horizontal speed 6 m/sec as shown in figure. If $m \ll M$, then for one dimensional elastic collision, the speed of lighter particle after collision will be:



- (1) 2 m/sec in original direction
(2) 2 m/sec opposite to the original direction
(3) 4 m/sec opposite to the original direction
(4) 4 m/sec in original direction
38. A block of mass $2M$ is held at rest when the spring is at its natural length as shown. Now the block is released suddenly in this position to fall vertically. Find the maximum elongation in the spring. (Spring is ideal)



- (1) $\frac{2Mg}{K}$ (2) $\frac{4Mg}{K}$
(3) $\frac{Mg}{K}$ (4) $\frac{Mg}{2K}$
39. The potential energy of a particle varies with distance x as shown in the graph. The force acting on the particle is zero at.

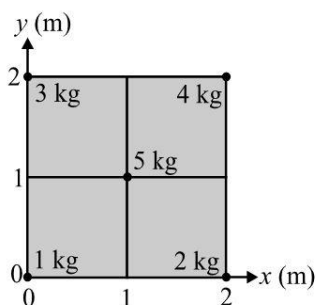


- (1) C (2) B
(3) B and C (4) A and D
40. Semicircular ring and the semicircular disc are cut from a circular ring and disc respectively, of radius R as shown in the figure. If the masses of the semicircular ring and semicircular disc are equal. The ratio of distance of center of mass of ring from point O and distance of center of mass of disc from point O', $\frac{x_1}{x_2}$ is:

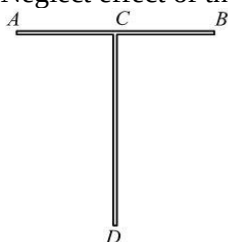


- (1) 1 : 1 (2) 3 : 2
(3) 3 : 4 (4) 4 : 5

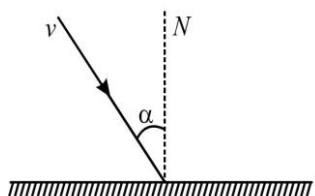
41. Five masses are placed in a plane as shown in the figure. The co-ordinates of the centre of mass are nearest to:



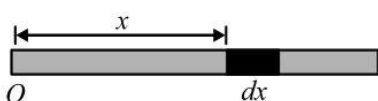
- (1) 1.2, 1.4 (2) 1.3, 1.1
(3) 1.1, 1.3 (4) 1.0, 1.0
42. Two identical thin uniform rods of length L each are joined to form T shape as shown in the figure. The distance of center of mass of resultant shape from D is: (Neglect effect of thickness)



- (1) 0 (2) $\frac{L}{4}$
(3) $\frac{3L}{4}$ (4) L
43. A ball of mass m hits a floor with a speed v making an angle α with the normal N . The coefficient of restitution is e . The angle made by reflected ball with normal is:



- (1) $\cot^{-1}(e \cos \alpha)$ (2) $\tan^{-1}(e \cot \alpha)$
(3) $\tan^{-1}\left(\frac{\tan \alpha}{e}\right)$ (4) $\cot^{-1}\left(\frac{\tan \alpha}{e}\right)$
44. The linear mass density of a straight rod of length L varies as $\lambda = A + Bx$, where x is the distance from the left end. The center of mass of rod from point O is:

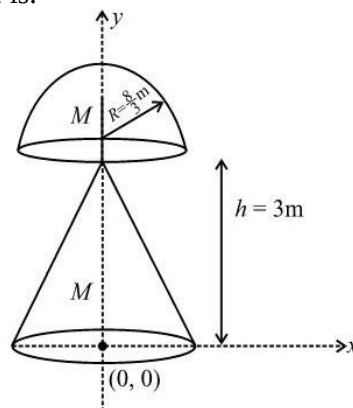


- (1) $\frac{AL + 2BL^2}{(2A + BL)}$ (2) $\frac{2AL + 2BL^2}{3(2A + BL)}$
(3) $\frac{3AL + 2BL^2}{2(2A + BL)}$ (4) $\frac{3AL + 2BL^2}{3(2A + BL)}$

45. A sphere of mass m moving with a constant velocity collides with another stationary sphere of same mass. The ratio of velocities of two spheres after collision will be, if the coefficient of restitution is e :

- (1) $\frac{1-e}{1+e}$ (2) $\frac{e-1}{e+1}$
(3) $\frac{e+1}{e-1}$ (4) 1

46. The co-ordinates of center of mass of system of solid hemisphere and hollow cone as shown in the figure is:



- (1) $\left(\frac{5}{2}, 0\right)$ (2) $\left(0, \frac{5}{2}\right)$
(3) $\left(\frac{3}{2}, 0\right)$ (4) $\left(0, \frac{3}{2}\right)$
47. The potential energy U varies with the position x of a particle according to relation $U(x) = (x^2 - 6x)$ J (where x is in meter). Choose the **correct** option.

- (1) $x = 3$ is a point of unstable equilibrium.
(2) $x = -2$ is a point of stable equilibrium.
(3) $x = 2$ is a point of equilibrium.
(4) $x = 3$ is a point of stable equilibrium.

48. Two similar spring P and Q have spring constants K_P and K_Q , such that $K_P > K_Q$. They are stretched first by the same amount (case a), then by the same force (case b). The work done by the springs W_P and W_Q are related as, in case (a) and case (b) respectively.

- (1) $W_P > W_Q$; $W_Q > W_P$
(2) $W_P < W_Q$; $W_Q < W_P$
(3) $W_P = W_Q$; $W_P > W_Q$
(4) $W_P = W_Q$; $W_P = W_Q$

49. Which of the following forces is called a conservative force?

- (1) Frictional force
(2) Air resistance
(3) Electrostatic force
(4) Viscous force

50. A force $\vec{F} = (6\hat{i} - 8\hat{j})N$, acts on a particle and displaces it over 4m along the positive X-axis and 6m along the positive Y-axis. The work done during the total displacement is;

- (1) 72 J (2) -24 J
(3) 24 J (4) Zero

SECTION-(II) CHEMISTRY**SECTION - A**

51. 1 mol each of the following compounds is dissolved in 1 L of solution. Which will have the largest ΔT_b value?
 (1) HF (2) HCl
 (3) HBr (4) HI
52. A solution of two volatile components showing negative deviation from Raoult's Law shows;
 (1) decrease in vapour pressure and increase in boiling point.
 (2) increase in vapour pressure and decrease in boiling point.
 (3) decrease in vapour pressure and decrease in boiling point.
 (4) increase in vapour pressure and increase in boiling point.
53. The solution having higher osmotic pressure than the reference solution is called;
 (1) hypertonic solution
 (2) isotonic solution
 (3) hypotonic solution
 (4) ideal solution
54. The gas which is expected to have maximum value of Henry's constant (K_H) will be;
 (1) BH_3 (2) CO_2
 (3) H_2S (4) He
55. For a very dilute solution of H_3PO_3 , Van't Hoff factor (i) is (Degree of dissociation (α) = 1);
 (1) $i = 7$ (2) $i = 3$
 (3) $i = 4$ (4) $i = 5$
56. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: The molarity of a solution changes with temperature.
Reason R: The volume of a solution changes with change in temperature.
 In the light of the above statements, choose the **correct** answer from the options given below.
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.
57. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: When methyl alcohol is added to water, the boiling point of water decreases.
Reason R: When a volatile solute is added to a volatile solvent, elevation in boiling point is observed.
 In the light of the above statements, choose the **correct** answer from the options given below.
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.
58. Which pair will have similar geometry?
 (1) Carbon dioxide and sulphur dioxide
 (2) Carbonate and sulphate ion
 (3) Ammonia and BH_3
 (4) Sulphate and perchlorate ion
59. An element has $[Ar]3d^1$ configuration in its +2 oxidation state. Its position in the periodic table is;
 (1) period-3, group-3
 (2) period-3, group-7
 (3) period-4, group-3
 (4) period-3, group-9
60. If $1/2$ moles of oxygen combine with aluminium to form $1/3$ mole of Al_2O_3 , then weight of aluminium metal used in the reaction is; (Molar atomic mass of Aluminium = 27 g/mol);
 $4Al(s) + 3O_2(g) \rightarrow 2Al_2O_3(s)$
 (1) 27 g (2) 18 g
 (3) 54 g (4) 40.5 g
61. Which of the following compound shows optical isomerism?
 (1) $CH_3CH_2CH_3$
 (2) $CH_2OHCHOHCH_2OH$
 (3) $CH_3CHOHC_2H_5$
 (4) CCl_2F_2
62. In the molecular orbital diagram for the molecular ion, N_2^+ , the number of electrons in the $\sigma 2p_z$ molecular orbital is:
 (1) 0 (2) 2
 (3) 3 (4) 1
63. 0.5 mole of H_2SO_4 is mixed with 0.2 mole of $Ca(OH)_2$. The maximum number of moles of $CaSO_4$ formed is:
 (1) 0.2 (2) 0.5
 (3) 0.4 (4) 1.5

64. **Correct** decreasing order of dipole moment is:-

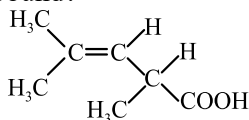
- (1) $\text{BF}_3 > \text{NF}_3 > \text{NH}_3$
- (2) $\text{NF}_3 > \text{BF}_3 > \text{NH}_3$
- (3) $\text{NH}_3 > \text{NF}_3 > \text{BF}_3$
- (4) $\text{NH}_3 > \text{BF}_3 > \text{NF}_3$

65. Arrange the following in the decreasing order of acidic strength:

- (A) $\text{NO}_2\text{CH}_2\text{COOH}$
- (B) $\text{NC-CH}_2\text{COOH}$
- (C) CCl_3COOH
- (D) CHCl_2COOH

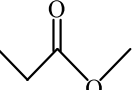
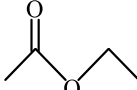
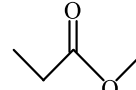
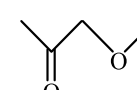
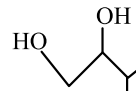
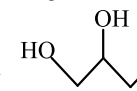
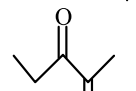
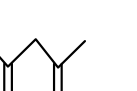
- (1) $\text{A} > \text{B} > \text{C} > \text{D}$ (2) $\text{C} > \text{D} > \text{A} > \text{B}$
- (3) $\text{B} > \text{A} > \text{D} > \text{C}$ (4) $\text{C} > \text{A} > \text{B} > \text{D}$

66. Which of the following isomerism is shown by the given compound?



- (1) Tautomerism
- (2) Optical isomerism
- (3) Geometrical isomerism
- (4) Geometrical and optical isomerism

67. Which of the following pairs of compounds are functional isomers?

- (1)  and 
- (2)  and 
- (3)  and 
- (4)  and 

68. Which of the following is **not** an electrophile?

- (1) CH_3^+ (2) AlCl_3
- (3) SiF_4 (4) $\left[(\text{CH}_3)_4\text{N}^+\right]^\ominus\text{OH}^-$

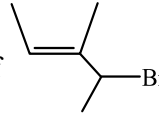
69. The ionic radii (in Å) of N^{3-} , O^{2-} and F^- are respectively;

- (1) 1.71, 1.40 and 1.36
- (2) 1.71, 1.36 and 1.40
- (3) 1.36, 1.40 and 1.71
- (4) 1.36, 1.71 and 1.40

70. Decreasing order of stability of given carbocations is as:

- (i)  (ii) $\text{CH}_3-\text{CH}_2^+$
- (iii) $\text{C}_6\text{H}_5-\text{CH}_2^+$ (iv) CH_3^+

- (1) $\text{iii} > \text{ii} > \text{iv} > \text{i}$
- (2) $\text{i} > \text{iii} > \text{iv} > \text{ii}$
- (3) $\text{i} > \text{iii} > \text{ii} > \text{iv}$
- (4) $\text{iii} > \text{ii} > \text{i} > \text{iv}$

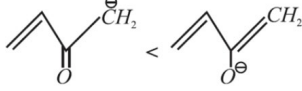
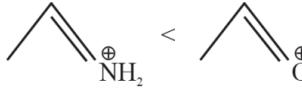
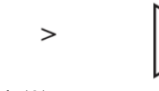
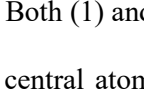
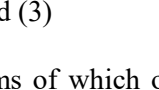
71. The IUPAC name of  is:

- (1) 2-Bromo-3-methylbut-3-ene
- (2) 4-Bromo-3-methylpent-2-ene
- (3) 2-Bromo-3-methylpent-3-ene
- (4) 4-Bromo-2, 3-dimethylbut-2-ene

72. The compound with zero dipole moment is;

- (1) cis-But-2-ene
- (2) trans-But-2-ene
- (3) But-1-ene
- (4) 2-Methylprop-1-ene

73. Choose the **correct** comparison of stability:

- (1)  < 
- (2)  < 
- (3)  > 
- (4) Both (1) and (3)

74. The central atoms of which one of the following molecules, has 3 bond pairs and 2 lone pairs of electrons.

- (1) Boron trifluoride
- (2) Phosphorous trichloride
- (3) Chlorine trifluoride
- (4) Xenon trioxide

75. What is the mole fraction of solvent in aqueous solution of NaOH having concentration 3 m?

- (1) 0.3 (2) 0.95
- (3) 0.7 (4) 0.05

76. The molarity of a 100 mL solution containing 5.1 g of hydrogen peroxide is;

- (1) 0.15 M (2) 1.5 M
- (3) 3.0 M (4) 50.0 M

77. Number of H-atoms in 0.046 g of ethanol is:

- (1) 6×10^{20} (2) 1.2×10^{21}
- (3) 3×10^{21} (4) 3.6×10^{21}

78. The percentage abundances of ^{12}C and ^{13}C are 98.9 and 1.1 respectively. The average atomic mass of carbon (in a.m.u) is;

- (1) 12.11 amu (2) 12.981 amu
- (3) 12.011 amu (4) 12.891 amu

79. Among the following, the aromatic compound is;

- (1)  (2) 
- (3)  (4) 

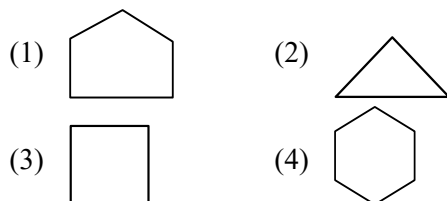
80. Match List-I and List-II.

	List-I (Compounds)		List-II (Geometry/Shape)
(A)	XeF ₆	(I)	Distorted octahedral
(B)	XeO ₃	(II)	Square planar
(C)	XeOF ₄	(III)	Pyramidal
(D)	XeF ₄	(IV)	Square pyramidal

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
- (2) A-I, B-II, C-IV, D-III
- (3) A-IV, B-III, C-I, D-II
- (4) A-IV, B-I, C-II, D-III

81. Which of the following is planar and **cannot** form a conformational isomer?



82. Strength of the hydrogen bond is determined by interaction between the

- I. lone pair of the electronegative atom of one molecule and the hydrogen atom of other molecule which is directly linked to electronegative element such as (F, O or N)
 - II. bond pair of the electronegative atom of one molecule and the hydrogen atom of other molecule.
- (1) Only I
 - (2) Only II
 - (3) Both I and II
 - (4) Neither I nor II

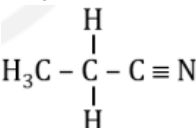
83. Which of the following pairs of compounds are chain isomers?

- (1) Isobutyl alcohol and sec-pentyl alcohol
- (2) Isobutyl alcohol and tert-butyl alcohol
- (3) sec-pentyl alcohol and neopentyl alcohol
- (4) Ethyl alcohol and ethylene glycol

84. Maximum number of lone pairs present among the following chemical species is:

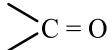
- (1) BF₄⁻ (2) BH₄⁻
- (3) CN⁻ (4) COCl₂

85. Which of the following represents the given mode of hybridization sp³, sp², sp² and sp from left to right?

- (1) H₂C = CH - C ≡ N
- (2) CH₃CH = CHCN
- (3) 
- (4) CH₂=CH-CH₂-NH₂

SECTION - B

86. The **correct** decreasing order of priority of functional groups in naming an organic compound as per the IUPAC system of nomenclature is:

- (1) -COOH > -CONH₂ > -COCl > -CHO
- (2) -SO₃H > -COCl > -CONH₂ > -CN
- (3) -COOR > -COCl > -NH₂ > 
- (4) -COOH > -COOR > -CONH₂ > -COCl

87. The ebullioscopic constant of a liquid solvent is the elevation of boiling point of;

- (1) one molar solution of non-volatile and non-electrolyte solute in it.
- (2) one normal solution of non-volatile and non-electrolyte solute in it.
- (3) one formal solution of non-volatile and non-electrolyte solute in it.
- (4) one molal solution of non-volatile and non-electrolyte solute in it.

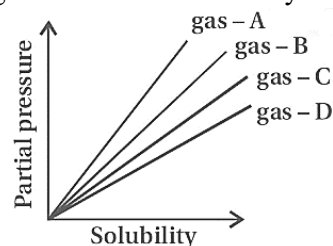
88. There are some of the characteristics of the supersaturated solution

- A. Equilibrium exists between solutions and solute.
- B. If a crystal of solute is added to supersaturated solution, crystallization occurs rapidly.
- C. Supersaturated solutions contain more solute than they should at a particular temperature.

Correct characteristics of supersaturated solutions are:

- (1) A, B, C (2) B, C
- (3) A, C (4) A, B

89. From the given graph at constant temperature, which gas has the least solubility?



- (1) gas-D (2) gas-B
- (3) gas-A (4) gas-C

90. The osmotic pressure of 5% (w/v) solution of urea at 27°C is

- (1) 20.5 atm (2) 10.5 atm
- (3) 12.5 atm (4) 15.5 atm

91. The plot of osmotic pressure (π) vs concentration (mol L⁻¹) for solution gives a straight line with slope 25.73 L bar mol⁻¹. The temperature at which the osmotic pressure measurement is done is:

- (use R = 0.083 L bar mol⁻¹ K⁻¹)
- (1) 12.05°C (2) 37°C
 - (3) 310°C (4) 25.73°C

92. Consider the following pairs of solution which will be isotonic at the same temperature, the number of pairs of solutions is/are _____
- A. 1 M aq. NaCl and 2 M aq. Urea
 B. 1 M aq. CaCl_2 and 1.5 M aq. KCl
 C. 1.5 M aq. AlCl_3 and 2 M aq. Na_2SO_4
 D. 2.5 M aq. KCl and 1 M aq. $\text{Al}_2(\text{SO}_4)_3$
- (1) A only (2) B and C only
 (3) A, C and D only (4) A, B, C and D

93. Match List-I with List-II.

List-I (Species)	List-II (Property)
(A) SO_2Cl_2	(I) Paramagnetic
(B) NO	(II) Diamagnetic
(C) NO_2^-	(III) Tetrahedral
(D) I_3^-	(IV) Linear

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
 (2) A-IV, B-I, C-III, D-II
 (3) A-II, B-III, C-I, D-IV
 (4) A-III, B-I, C-II, D-IV

94. Match List-I with List-II.

List-I (Atomic number of elements)	List-II (Position of elements in periodic table)
(A) 19	(I) p-block
(B) 22	(II) f-block
(C) 32	(III) d-block
(D) 64	(IV) s-block

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
 (2) A-IV, B-III, C-I, D-II
 (3) A-IV, B-I, C-III, D-II
 (4) A-II, B-I, C-III, D-IV

95. Match List-I with List-II.

List-I (Property)	List-II (Molecule)
(A) Resonance	(I) HF
(B) Hydrogen bonds	(II) C
(C) Covalent solid	(III) LiF
(D) Ionic solid	(IV) O_3

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
 (2) A-IV, B-I, C-III, D-II
 (3) A-IV, B-I, C-II, D-III
 (4) A-III, B-IV, C-II, D-I

96. Match List-I with List-II.

List-I (Electronic configuration)	List-II (Electron gain enthalpy/ kJ mol^{-1})
(A) $1s^2 2s^2 2p^6$	(I) -53
(B) $1s^2 2s^2 2p^6 3s^1$	(II) -328
(C) $1s^2 2s^2 2p^5$	(III) -141
(D) $1s^2 2s^2 2p^4$	(IV) +116

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-II, D-III
 (2) A-III, B-II, C-II, D-IV
 (3) A-IV, B-III, C-I, D-II
 (4) A-IV, B-I, C-III, D-II

97. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Pent-1-ene and pent-2-ene are position isomers.

Reason R: Position isomers differ in the position of functional group or a substituent.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.

98. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Atomicity of oxygen is 2.

Reason R: 1 mole of an element contains 6.023×10^{23} atoms

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.

99. Given below are two statements.

Statement I: The balancing of chemical equations is based on law of conservation of mass.

Statement II: Total mass of reactants is equal to total mass of products.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

100. This question consists of two statements. Choose the **correct** option.

Statement I: Equal volumes of all gases under similar conditions of temperature and pressure contain equal number of molecules.

Statement II: Volume of a gas is inversely proportional to pressure.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

SECTION-(III) BOTANY**SECTION - A****101.** Select the **correct** option for polysome.

- (1) Polysome are present in eukaryotes.
- (2) They are about 15 mm by 20 mm in size.
- (3) They translate the proteins into mRNA.
- (4) They are also known as polyribosomes.

102. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Disintegration of nuclear membrane	(I)	Anaphase
(B)	Appearance of nucleolus	(II)	Prophase
(C)	Division of centromere	(III)	Telophase
(D)	Synthesis of RNA and proteins	(IV)	G ₂ -phase

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
- (2) A-II, B-IV, C-III, D-I
- (3) A-II, B-III, C-I, D-IV
- (4) A-I, B-II, C-III, D-IV

103. Identify the diagram and select the **correct** shape for the indicated cell type..

- A. White blood cells
- B. Red blood cells
- C. Round and biconcave
- D. Amoeboid
- E. Round and oval

Choose the most appropriate answer from the options given below:

- (1) B and E only
- (2) A and D only
- (3) A, D and E only
- (4) D and E only

104. Select **correct** option for nucleolus.

- (1) Rounded structure found in cytoplasm near nucleus.
- (2) Spherical structure inside nucleus and having rRNA.
- (3) Rod-shaped structure in cytoplasm near the nucleus.
- (4) Sausage structure in cytoplasm near the nucleus.

105. Select the **incorrect** option for vacuole.

- (1) Contractile vacuoles help in osmoregulation.
- (2) Food vacuoles are formed by engulfing the food particles.
- (3) The vacuole is the membrane bound space found in the cytoplasm.
- (4) Gas vacuoles occur in unicellular eukaryote.

106. Given below are two statements:

Statement I: Microtubules, macrofilaments and intermediate filaments are collectively referred to as the cytoskeleton.

Statement II: Cytoskeleton in a cell helps in motility.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

107. Select the **correct** statement regarding bivalent.

- (1) Bivalent are tetrads.
- (2) A bivalent means 2 chromatids and 2 centromere.
- (3) One bivalent consists of 4 homologous chromosomes.
- (4) Bivalents formation occurs in leptotene.

108. If the initial amount of DNA is denoted as 2C and chromosome number 2n then after S-phase:

- (1) DNA content is 4C and chromosome number 4n.
- (2) DNA content is 4C and chromosome number 8n.
- (3) DNA content is 2C and chromosome number 4n.
- (4) DNA content is 4C and chromosome number 2n.

109. Consider the following taxonomic categories and find out which can be common for *Datura* and *Triticum*.

- A. Division
- B. Class
- C. Kingdom
- D. Genus

Choose the most appropriate answer from the options given below:

- (1) A and D only
- (2) A and C only
- (3) A, B and C only
- (4) A, B, C and D

110. The **correct** sequence of phases of cell cycle is:

- (1) M→G₁→G₂→S.
- (2) G₁→G₂→S→M.
- (3) S→G₁→G₂→M.
- (4) G₁→S→G₂→M.

111. Recombination nodules are formed before:

- (1) leptotene. (2) zygotene.
(3) pachytene. (4) diplotene.

112. Cells which are **not** dividing are likely to be at:

- (1) Gap 1 phase. (2) Gap 2 phase.
(3) quiescent stage. (4) synthesis phase.

113. Given below are two statements:

Statement I: In the S and G₂ phases the new DNA molecules formed are not distinct but intertwined.

Statement II: Metaphase chromosome is made up of two sister chromatids.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

114. Consider the following statements and select the **incorrect** statement/s.

- A. Centrosome undergoes duplication during interphase.
B. In plant cells, wall formation starts at the centre which grows outwards.
C. In animal cells, during the S phase, DNA replication begins in the nucleus.
D. The chromosome number increase after M phase.

Choose the most appropriate answer from the options given below:

- (1) D only (2) A and B only
(3) A, B and C only (4) A, B, C and D

115. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Deuteromycetes are commonly known as imperfect fungi.

Reason R: Once asexual stages of members of deuteromycetes were discovered they were often moved to ascomycetes and basidiomycetes.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

116. Select the **correct** option for **P** and **Q** respectively by studying the following table which shows different organisms with their taxonomic categories.

Common Name	Order	Class
Dog	Carnivora	P
Cat	Q	Mammalia

P

- (1) Mammalia
(2) Chordata
(3) Mammalia
(4) Chordata

Q

- Diptera
Carnivora
Carnivora
Primata

117. Consider the following statements.

- A. Numerical taxonomy is based on all non-observable characteristics.
B. Cytotaxonomy is based on cytological information like chloroplast number.
C. Phylogenetic classification systems based on evolutionary relationships between the various organisms.
D. Chemotaxonomy that uses the chemical constituents of the plant.

Choose the most appropriate answer from the options given below:

- (1) A and D are correct only
(2) A and B are incorrect only
(3) B and D are incorrect only
(4) A, B, C and D are correct

118. Select the **incorrect** option from the following.

- (1) Bacteria are simple in behaviour.
(2) Compared to many other organisms, bacteria as a group show the most extensive metabolic diversity.
(3) Bacteria are the sole members of the Kingdom Monera.
(4) Bacterial structure is very simple.

119. Asexual reproduction in algae occurs by:

- (1) isogamy and anisogamy only.
(2) spores only.
(3) oogamy and anisogamy.
(4) isogamy, anisogamy and oogamy.

120. Select the **incorrect** statement.

- (1) Biological names are generally in Latin.
(2) Binomial nomenclature has two components.
(3) The words in a biological name, when handwritten, are separately underlined.
(4) Both the words in biological names start with a capital letter.

121. Given below are two statements:

Statement I: The word systematics is derived from the Greek word 'systema'.

Statement II: 'Systema' means systematic arrangement of organisms.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

122. Fusion of two nuclei is called:

- (1) karyogamy only.
- (2) plasmogamy only.
- (3) plasmodium.
- (4) karyogamy and plasmogamy.

123. Select the **incorrect** match for mitochondria from the following.

(1)	Matrix	DNA
(2)	Cristae	Outer membrane
(3)	Shape	Sausage
(4)	Division	Fission

124. Choose the **correct** statement.

- (1) Golden algae are multicellular eukaryotes.
- (2) *Plasmodium* has gullet.
- (3) *Gonyaulax* is golden algae.
- (4) Spores in slime moulds are dispersed by air currents.

125. Identify the fungi which belongs to the group of sac fungi:

- (1) mushrooms. (2) puffballs.
- (3) yeast. (4) bracket fungi.

126. Phycoerythrin is the major pigment in:

- (1) red algae. (2) green algae.
- (3) brown algae. (4) bryophytes.

127. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The number of species that are known and described range between 1.7-1.8 billion.

Reason R: Biodiversity is the number and types of organisms present on earth.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

128. Identify the **incorrect** match from the following.

- (1) *Puccinia* → Rust fungus
- (2) Yeast → Bread and beer
- (3) Chrysophyte → Diatomaceous earth
- (4) *Penicillium* → Smut fungus

129. In oogamy:

- (1) Female gamete is flagellate.
- (2) Male gamete and female gamete are non-flagellate.
- (3) Male gamete is small.
- (4) Male gamete is non-motile.

130. ICZN stands for:

- (1) Indian Code of Zoological Names.
- (2) Indian Code of Zoological Nomenclature.
- (3) International Code of Zoological Names.
- (4) International Code of Zoological Nomenclature.

131. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Amyloplasts	(I)	Carotene
(B)	Elaioylasts	(II)	Protein
(C)	Aleuroplasts	(III)	Starch
(D)	Chromoplasts	(IV)	Oils and fat

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-IV, B-II, C-I, D-III
- (3) A-III, B-IV, C-II, D-I
- (4) A-II, B-I, C-III, D-IV

132. At least a half of the total CO₂ fixation on earth is carried out through photosynthesis by:

- (1) protista. (2) monera.
- (3) algae. (4) bryophytes.

133. The major pigments in green algae are A and B and stored food is C. Identify **A**, **B** and **C** respectively.

	A	B	C
(1)	chl <i>a</i>	chl <i>d</i>	starch
(2)	chl <i>a</i>	chl <i>c</i>	floridean starch
(3)	chl <i>a</i>	chl <i>b</i>	starch
(4)	chl <i>a</i>	chl <i>b</i>	mannitol

134. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The main plant body of the bryophyte is haploid.

Reason R: The main plant body of the bryophyte is sporophyte.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

135. Each genus may have one or more than one:

- (1) specific epithets. (2) kingdom.
- (3) class. (4) phylum.

SECTION - B

136. Given below are two statements:

Statement I: The plant body of pteridophyte is attached to the substratum by unicellular rhizoids.

Statement II: The leaves in pteridophyta are small (microphylls) as in *Selaginella*.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

137. Of the total duration of cell cycle, the interphase lasts for more than:

- (1) 95%. (2) 5%.
(3) 90%. (4) 80%.

138. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Diakinesis is the final stage of prophase-I.

Reason R: Terminalisation of chiasmata occurs in diakinesis.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

139. Match List-I with List-II:

List-I		List-II	
(A)	Zoospores	(I)	Asexual spores in deuteromycetes and ascomycetes
(B)	Aplanospores	(II)	Motile asexual spores in phycomycetes
(C)	Conidia	(III)	Non-motile asexual spores in phycomycetes
(D)	Basidiospores	(IV)	Sexual spores in basidiomycetes

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
(2) A-II, B-III, C-I, D-IV
(3) A-III, B-I, C-IV, D-II
(4) A-I, B-II, C-IV, D-III

140. Algae are largely:

- (1) saprophytic. (2) terrestrial.
(3) parasite. (4) aquatic.

141. Consider the following statements.

- A. Centrosome is an organelle.
B. Centrioles are spherical structure.
C. Centrosome contain two centrioles.
D. Centrioles in a centrosome lie parallel to each other.

Choose the most appropriate answer from the options given below:

- (1) A and D are correct only
(2) A and B are incorrect only
(3) B and D are incorrect only
(4) A, B, C and D are correct

142. Two equatorial plates are formed during which phase of cell cycle?

- (1) Metaphase-II (2) Metaphase-I
(3) Interphase (4) Metaphase of Mitosis

143. In which of the following, all listed genera belong to the same class of algae.

- (1) *Chara*, *Fucus*, *Polysiphonia*
(2) *Volvox*, *Spirogyra*, *Chlamydomonas*
(3) *Porphyra*, *Ectocarpus*, *Ulothrix*
(4) *Sargassum*, *Laminaria*, *Gracilaria*

144. Identify the algae which produces algin.

- (1) Green algae (2) Blue-green algae
(3) Brown algae (4) Red algae

145. Select the **incorrect** statement for viroids.

- (1) Discovered in 1971.
(2) Viroids do not have capsid.
(3) Its protein coat is same as virus.
(4) These were discovered by T.O Diener.

146. Identify from the following which belong to the genus *Panthera*.

- A. Cat B. Lion
C. Leopard D. Dog

Choose the most appropriate answer from the options given below:

- (1) D only (2) A and B only
(3) B and C only (4) A, B, C and D

147. In the hierarchy of classification order is present between:

- (1) family and genus.
(2) phylum and kingdom.
(3) family and species.
(4) family and class.

148. The criteria used by Whittaker for classification:

- A. Cell structure
B. Body organisation
C. Mode of nutrition
D. Habitat
E. Phylogenetic relationships

Choose the most appropriate answer from the options given below:

- (1) A, C and D only (2) A and C only
(3) B, C and D only (4) A, B, C and E only

149. Golgi complex and lysosome are:

- (1) involved in protein synthesis.
(2) responsible for secretion.
(3) part of endomembrane system.
(4) formed by vacuole.

150. The pili in bacteria:

- (1) are round structures.
(2) made up of carbohydrate.
(3) are elongated tubular structures.
(4) helps in respiration.

SECTION-(IV) ZOOLOGY**SECTION - A****151.** Match the **List-I** with **List-II**.

List-I		List-II	
A.	Cuboidal epithelium	I.	Wall of intestine
B.	Ciliated epithelium	II.	Moist surface of buccal cavity
C.	Compound epithelium	III.	Fallopian tube
D.	Columnar epithelium	IV.	Ducts of glands

Choose the **correct** answer from the option given below.

- (1) A-IV; B-III; C-II; D-I
- (2) A-III; B-II; C-I; D-IV
- (3) A-I; B-III; C-II; D-IV
- (4) A-III; B-IV; C-I; D-II

152. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): Skeletal muscle fibres contract in response to stimulation, then relax and return to their uncontracted state in an uncoordinated fashion.

Reason (R): Their action moves the body to adjust to the changes in the environment and to maintain the positions of the various parts of the body.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

153. Which of the following options is **correct** about ERV (expiratory reserve volume)?

- (1) Its value is less than TV (tidal volume).
- (2) It is the additional volume of air, a person can expire by a forcible expiration.
- (3) $ERV = EC - TV$, where EC = Expiratory capacity and TV is Tidal volume.
- (4) Both (2) and (3)

154. Read the following statements (**I–IV**).

- I. The relaxation of external intercostal muscles lifts up the ribs and sternum.
- II. The thoracic cavity is anatomically an air tight chamber.
- III. Healthy man can inspire approximately 200 ml of air per minute.
- IV. During expiration, the intrapulmonary pressure is slightly below the surrounding atmospheric pressure.

Choose the option containing **incorrect** statement/s.

- (1) I and II only
- (2) III and IV only
- (3) I, III and IV only
- (4) I, II and III only

155. The development of *Periplanata americana* is:

- (1) holometabolous.
- (2) paurometabolous.
- (3) hemimetabolous.
- (4) both (1) and (3)

156. All of the following statements are correct **except**:

- (1) Frog is a poikilothermic animal.
- (2) Digestive enzymes are secreted by exocrine glands.
- (3) In male cockroaches, sperms are glued together in the form of bundles called spermatophores.
- (4) Salivary gland is a unicellular gland.

157. The excretory waste product in frog is:

- (1) ammonia.
- (2) urea.
- (3) uric acid.
- (4) all of these

158. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): Muscular tissues protect the under lying and over lying tissues.

Reason (R): Materials are exchanged at the surfaces across the epithelial tissues.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

159. Given below are two statements:

Statement I: $p\text{CO}_2$ of the alveoli is more than $p\text{CO}_2$ of atmospheric air.

Statement II: Solubility of O_2 is 20-25 times higher than the CO_2 .

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

160. The compound eyes of cockroaches consists of about _____ ommatidia.

Choose the option which fill the blanks **correctly**.

- (1) 200 hexagonal ommatidia
- (2) 200,000 hexagonal ommatidia
- (3) 20 hexagonal ommatidia
- (4) 2000 hexagonal ommatidia

161. Match the **List-I** with **List-II**.

List-I		List-II	
A.	Goblet cells	I.	Cluster of cells
B.	Adipose tissue	II.	Alimentary canal
C.	Endocrine gland	III.	Fat storage
D.	Salivary gland	IV.	Hormone secretion

Choose the **correct** answer from the option given below.

- (1) A-I, B-II, C-III, D-IV
- (2) A-II, B-IV, C-III, D-I
- (3) A-II, B-III, C-IV, D-I
- (4) A-I, B-II, C-IV, D-III

162. Enzyme required for joining of C-O, C-S, C-N, P-O, etc., bonds is;

- (1) ligase.
- (2) lyase.
- (3) polymerase.
- (4) hydrolase.

163. Which of the following statements is **incorrect** for epithelial tissue?

- (1) On the basis of structural modification of the cells, simple epithelium is divided into three types.
- (2) The cells are compactly packed with maximum intercellular matrix.
- (3) Free surface may have microvilli.
- (4) Specialised junctions are present.

164. Given below are two statements:

Statement I: In stone breaking industry, so much dust is produced that the defense mechanism of the body cannot fully cope with the situation.

Statement II: Binding of oxygen with haemoglobin is primarily related to partial pressure of O_2 .

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

165. In frog, _____ includes olfactory lobes, paired cerebral hemispheres and unpaired diencephalon.

Choose the option which fills the blank **correctly**.

- (1) forebrain
- (2) medulla
- (3) midbrain
- (4) cerebellum

166. Choose the **odd** pair out in the following.

- (1) Areolar connective tissue - Collagen fibres
- (2) Epithelium - Keratin
- (3) Neuron - Melanin
- (4) Muscle fibre - Actin

167. Chitinous exoskeleton is found in:

- (1) birds. (2) turtle.
- (3) insects. (4) fishes.

168. Which of the following is common feature of *Torpedo* and *Trygon*?

- (1) They live in fresh water.
- (2) They possess electric organs.
- (3) They have placoid scales and belong to class chondrichthyes.
- (4) They possess a poison sting on the tail.

169. Triglyceride consists of;

- (1) three fatty acids + two glycerols
- (2) one fatty acid + three glycerols
- (3) one fatty acid + one glycerol
- (4) three fatty acids + one glycerol

170. Refer to the given below diagram and choose the **correct** option.

- I. Triploblastic and bilateral symmetry
- II. Metamerically segmented and coelomate animals
- III. Protonephridia
- IV. Closed circulatory system
- V. Dioecious



- (1) I, II and IV only
- (2) III, IV and V only
- (3) I, III, IV and V only
- (4) II, III and V only

171. In which of the following phylum body cavity is **not** lined by mesoderm?

- (1) Porifera
- (2) Ctenophora
- (3) Echinodermata
- (4) Both (1) and (2)

172. Which of the following animals comprises "Jawless fish"?

- (1) Lamprey & Scoliodon
- (2) Exocoetus & Labeo
- (3) Pterophyllum & Myxine
- (4) Petromyzon & Hagfish

173. Read the following features.

- (i) Bilateral symmetry
- (ii) Triploblastic
- (iii) Organ level of organisation
- (iv) Dorsoventrally flattened body

Choose the **correct** option from the above given characteristic.

- (1) Platyhelminthes
- (2) Aschelminthes
- (3) Annelida
- (4) Arthropoda

174. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): Aves and Mammals are homoiothermous.

Reason (R): They are unable to maintain a constant body temperature.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

175. True coelom appeared first in the course of evolution in:

- (1) aschelminthes.
- (2) chordata.
- (3) echinodermata.
- (4) annelida.

176. In which of the following organism, notochord extends from head to tail region and is persistent throughout their life?

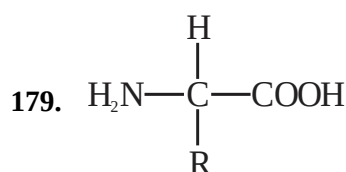
- (1) *Ascidia*
- (2) *Branchiostoma*
- (3) *Doliolum*
- (4) *Salpa*

177. Which of the following is **incorrect** for the class mammalia?

- (1) *Platypus* is oviparous.
- (2) Bats have wings.
- (3) Elephant is a viviparous.
- (4) Diaphragm is absent in *Macaca*.

178. Which of the following is **correctly** matched?

- (1) Radial symmetry - Coelenterates
- (2) Coelomates - Aschelminthes
- (3) Metamerism - Molluscs
- (4) Triploblastic - Sponges



is the general formula of amino acid. Here R stands for:

- (1) an amino group only.
- (2) a carboxylic group only.
- (3) a variable group.
- (4) a hydroxyl group only.

180. Each nucleotide is made up of;

- (1) nitrogen containing carbohydrate and pentose sugar with phosphate.
- (2) cyclic nitrogenous base + pentose sugar and a phosphate group.
- (3) nitrogenous base + hexose sugar + phosphate.
- (4) amino sugar + ribose + phosphate.

181. Which of the following is the **incorrect** statement?

- (1) Inorganic catalysts work efficiently at low temperature and low pressure while enzymes get damaged at low temperature (40°C).
- (2) Thermophilic organisms living in hot vents and sulphur springs have enzymes that are stable and retain their catalytic power even at high temperature (80 – 90°C).
- (3) Ribozymes are nucleic acids behaving like enzymes.
- (4) Malonate is a competitive inhibitor of succinate dehydrogenase.

- 182.** Which of the following is the **correct** statement w.r.t proteins?
- (1) Tertiary structure is necessary for biological activity of protein.
 - (2) They are homopolymers.
 - (3) It consists of only branched chains of amino acids.
 - (4) Dietary proteins are the source of non-essential amino acids.
- 183.** Which of the following type of bond is present between phosphate group and sugar in a nucleotide?
- (1) Peptide bond
 - (2) Ester bond
 - (3) Glycosidic bond
 - (4) Ether bond
- 184.** Select the **incorrect** statement.
- (1) Amino acids are substituents methanes.
 - (2) Glycerol is a trihydroxy propane.
 - (3) Lysine is a neutral amino acid.
 - (4) Lecithin is a phospholipid.
- 185.** Which of the following feature(s) is/are common to all those compounds found in acid soluble pool or filtrate?
- (1) They are called macromolecules or biomacromolecules.
 - (2) The molecular weight is more than ten thousand Daltons or above.
 - (3) They have molecular weight ranging from 18 to around 800 Da approximately.
 - (4) Both (1) and (2)

SECTION - B

186.

Pressure of gases	Systemic veins	Systemic arteries
O ₂	40 mm Hg	95 mm Hg
CO ₂	X	Y

Choose the **correct** option for **X** and **Y** to complete the given data.

- (1) **X** – 45mmHg, **Y** – 40mmHg
 - (2) **X** – 45 mmHg, **Y** – 45mmHg
 - (3) **X** – 45mmHg, **Y** – 50mmHg
 - (4) **X** – 45 mmHg, **Y** – 55mmHg
- 187.** Read the following characteristics.
- (i) It is located in pons.
 - (ii) It moderates the functions of respiratory rhythm centre.
 - (iii) Its absence results in an increase in depth of respiration.
- Choose the **correct** option w.r.t the above given characteristics.
- (1) Pneumotaxic centre.
 - (2) Respiratory rhythm centre.
 - (3) Chemosensitive zone.
 - (4) Thirst centre.

188. Read the following statements.

- I. Cell aggregate body plan is found in *Hydra*.
- II. Radial symmetry is the most common symmetry found in *Delphinus*.
- III. Pseudocoelom found in phylum Aschelminthes only.
- IV. All triploblastic animals have a true coelom.
- V. *Saccoglossus* exhibits internal fertilisation.

Which of the following option is **correct**?

- (1) I and V are true and II, III and IV are false.
- (2) II, III and V are true and I and IV are false.
- (3) I, II and III are true and IV and V are false.
- (4) I, II, IV and V are false, only III is true.

189. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion (A): Expiration takes place when the intra-pulmonary pressure is higher than the atmospheric pressure.

Reason (R): During expiration contraction of the diaphragm and the inter-costal muscles reduce the thoracic volume and thereby the pulmonary volume.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

190. In cockroach, the number of ganglia are:

- (1) two in thoracic and four in abdominal.
- (2) three in thoracic and six in abdominal.
- (3) three in thoracic and five in abdominal.
- (4) two in thoracic and six in abdominal.

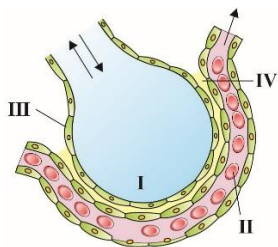
191. Exposure to carbon monoxide is harmful to animals because:

- (1) it reduces CO₂ transport.
- (2) it reduces O₂ transport.
- (3) it increases CO₂ transport.
- (4) it increases O₂ transport.

192. Choose the **incorrect** statement.

- (1) The thickness of diffusion membrane is less than a millimetre.
- (2) The human lungs comprise the branching network of bronchi, bronchioles and alveoli.
- (3) Pressure contributed by an individual gas in a mixture of gases is called partial pressure.
- (4) Aquatic arthropods exhibit pulmonary respiration.

193. Refer to the given below diagram and choose the option that represent the one celled thick alveolar wall.



- (1) I (2) II
(3) III (4) IV
194. Match the **List-I** with **List-II** w.r.t *Rana tigrina*.

List-I		List-II	
A.	Tongue	I.	10–12 in number
B.	Vasa efferentia	II.	Double fold of peritoneum
C.	Mesorchium	III.	Nucleated
D.	RBCs	IV.	Bilobed

Choose the **correct** answer from the option given below.

- (1) A-IV, B-I, C-II, D-III
(2) A-I, B-III, C-II, D-IV
(3) A-IV, B-II, C-III, D-I
(4) A-II, B-III, C-I, D-IV
195. When 1200 mL air is left in the lungs, it is called:
- (1) vital capacity.
(2) tidal volume.
(3) residual volume.
(4) inspiratory reserve volume.
196. All of the following statements are correct **except**:
- (1) Spices are secondary metabolites.
(2) GLUT-4 (glucose transporter 4) helps transport glucose into cells
(3) Coenzyme NADP contains vitamin thiamine
(4) Prosthetic group is tightly bound to apoenzyme.

197. Which of the following factors is/are favourable for the formation of oxyhaemoglobin?

- I. High pO_2 , low pCO_2
II. Lesser H^+ concentration, lower temperature
III. Low pO_2 , high pCO_2
IV. High H^+ , higher temperature
- (1) Only I (2) I and II only
(3) II and III only (4) III and IV only

198. Select the **odd** one out option.

- (1) *Corvus* (2) *Columba*
(3) *Bufo* (4) *Neophron*

199. Given below are two statements:

Statement I: Bioluminescence is the property of a living organism to emit light well-marked in *Taenia*.

Statement II: *Planaria* possess high regeneration capacity.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

200. Complete the analogy w.r.t the carbon atoms and choose the **correct** option.

Palmitic acid: 16 :: Arachidonic acid: ____

- (1) 20
(2) 22
(3) 24
(4) 18

